

Hierarchical Multi-Fidelity Inverse Design & Atomistic Validation of Multi-Principal Element Alloys

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Research Proposal: Hierarchical Multi-Fidelity Inverse Design and Atomistic Validation of Multi-Principal Element Alloys for Extreme Environments

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Abstract

Exploring the vast multi-component compositional space of Multi-Principal Element Alloys (MPEAs) and High-Entropy Alloys (HEAs) presents a formidable combinatorial challenge, rendering conventional empirical trial-and-error synthesis economically intractable. This proposal presents **MetaForge**, an Integrated Computational Materials Engineering (ICME) framework that couples physics-informed compositional screening with atomistic graph neural network (GNN) relaxations and a first-principles validation roadmap.

The framework implements a hierarchical multi-fidelity screening architecture:

- Tier 1 (High-Throughput Compositional Screening):** Enforces thermodynamic Hume-Rothery size mismatch ($\delta \leq 6.6\%$), Guo valence electron concentration (VEC) phase boundaries, Miedema binary mixing enthalpies (-15.0 to $+5.0$ kJ/mol), and Yang-Zhang thermodynamic parameters ($\Omega \geq 1.1$). Surrogates trained on 132-dimensional Matminer Magpie descriptors predict continuum density and dislocation-calibrated yield strengths based on the Taylor-Varvenne dislocation model: $\sigma_y = \sigma_0 + (M * \tau_{ss})$.
- Tier 2 (Atomistic SQS Annealing & GNN Potentials):** Special Quasirandom Structures (SQS, 54-atom BCC / 48-atom FCC) are synthesized by minimizing Warren-Cowley Short-Range Order (SRO) parameters via Monte Carlo simulated annealing, followed by structural relaxation using the universal CHGNet interatomic neural potential.
- Tier 3 (Quantum & Synthesis Roadmap):** Outlines ab-initio Density Functional Theory (DFT) calculations of ground-state elastic stiffness tensors (C11, C12, C44) and vacuum arc remelting synthesis.

Genetic algorithm optimization adhering strictly to the Yeh & Cantor multi-principal element criterion (5 to 35 atomic % per constituent) isolated a standout refractory HEA candidate: **W:18.5% - Mo:24.7% - Ta:26.1% - Nb:16.8% - V:13.9%**. This alloy exhibits a predicted yield strength of **1.96 GPa (1960 MPa)**, relaxed density of **14.29 g/cm³**, and a near-zero formation enthalpy ($\Delta E_f = +0.0286$ eV/atom = +2.76 kJ/mol) that is completely stabilized by high configurational entropy ($\Delta S_{mix} = 13.4$ J/mol·K). The interactive computational engine is deployed live at metaforge-web.onrender.com.

1. Background & Theoretical Framework

1.1 Multi-Principal Element Metallurgy

Unlike classical physical metallurgy—which optimizes properties by perturbing a single principal solvent lattice (e.g., Fe in steels, Ni in superalloys)—High-Entropy Alloys rely on four core metallurgical effects:

- **High Entropy Effect:** High configurational entropy of mixing ($\Delta S_{config} \geq 1.5 \cdot R$) lowers total Gibbs free energy ($\Delta G = \Delta H - T \cdot \Delta S$) at elevated temperatures, suppressing complex brittle intermetallics in favor of disordered solid solutions.
- **Sluggish Diffusion:** Fluctuating local atomic potential wells increase activation energies for vacancy migration, conferring exceptional high-temperature creep resistance.
- **Severe Lattice Distortion:** Size and modulus misfits among distinct constituents generate non-uniform internal stress fields, elevating intrinsic Peierls-Nabarro friction stress.
- **Cocktail Effect:** Non-linear synergistic interactions among constituents yield multi-functional mechanical and chemical enhancements.

1.2 The Combinatorial Bottleneck

Selecting 5 principal components from a candidate pool of 70 metallic elements yields over 12 million equiatomic alloy systems. Discretizing compositional variations by 5 atomic % increments expands the search hyper-volume to hundreds of millions of distinct candidates. High-throughput physical synthesis remains cost-prohibitive for broad sweeps. Informatics-guided inverse design is mandatory to down-select optimal candidates prior to laboratory synthesis.

2. Research Methodology & Multi-Fidelity Architecture

METAFORGE HIERARCHICAL WORKFLOW

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[Tier 1] Combinatorial Screening (Simplex Discretization)
• Hume-Rothery Lattice Distortion Filter: $\delta \leq 6.6\%$
• Guo Valence Electron Concentration (VEC) Phase Stability
• Miedema Enthalpy of Mixing: $-15.0 \leq \Delta H_{\text{mix}} \leq +5.0 \text{ kJ/mol}$
• Yang & Zhang Thermodynamic Ratio: $\Omega \geq 1.1$
• Featurization: 132 Magpie Descriptors (Matminer)
• Surrogate Property Modeling:
 - Continuum Density (g/cm^3)
 - Dislocation Yield Strength: $\sigma_y = \sigma_0 + M * \tau_{ss}$

[Tier 2] Atomistic Generation & Neural Potential Relaxation
• SQS Synthesis: Warren-Cowley SRO Minimization (Monte Carlo)
• 54-atom BCC / 48-atom FCC Supercell Geometry
• Universal Interatomic Potential Relaxation (CHGNet)
• Formation Energy: $\Delta E_f = E_{\text{relaxed}}/N - \sum(c_i * E_{\text{ref}})$

[Tier 3] Validation & Laboratory Synthesis Roadmap
• Ab-Initio Quantum DFT (VASP / Quantum ESPRESSO)
• Elastic Stiffness Tensors (C11, C12, C44) & Moduli
• Vacuum Arc Remelting, XRD Phase Analysis, Vickers Microhardness
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### 2.1 Thermodynamic Phase Stability & Solid-Solution Criteria

To ensure that predicted alloys stabilize as disordered single-phase solid solutions, candidate compositions must satisfy four coupled thermodynamic criteria:

1. ATOMIC SIZE MISMATCH ( $\delta$ ):  
$$\delta = 100 * \text{SquareRoot}(\text{Sum of } [c_i * (1 - r_i / r_{\text{average}})^2]) \leq 6.6\%$$
where  $r_i$  is atomic radius and  $r_{\text{average}}$  is average radius.
2. MIEDEMA ENTHALPY OF MIXING ( $\Delta H_{\text{mix}}$ ):  
$$\Delta H_{\text{mix}} = \text{Sum of } [4 * H_{ij} * c_i * c_j]$$
Enforced Range:  $-15.0 \text{ kJ/mol} \leq \Delta H_{\text{mix}} \leq +5.0 \text{ kJ/mol}$ .
3. YANG-ZHANG THERMODYNAMIC RATIO ( $\Omega$ ):  
$$\Omega = (\text{Average Melting Point} * \Delta S_{\text{config}}) / |\Delta H_{\text{mix}}| \geq 1.1$$
4. VALENCE ELECTRON CONCENTRATION (VEC):  
$$\text{VEC} = \text{Sum of } [c_i * \text{VEC}_i]$$
  - $\text{VEC} < 6.87 \implies$  Pure Body-Centered Cubic (BCC)
  - $\text{VEC} \geq 8.0 \implies$  Pure Face-Centered Cubic (FCC)

### 2.2 Yield Strength Formulation (Dislocation Theory)

Rather than treating yield strength as an empirical proxy, MetaForge implements a physical dislocation model based on the Taylor factor and Varvenne-Luque-Curtin solid-solution strengthening:

EQUATION: Dislocation Yield Strength  
 $\sigma_y = \sigma_0 + [ M * \tau_{ss} ]$

Where:

- $\sigma_0$  = Intrinsic Peierls-Nabarro friction stress of pure lattice (approx.  $G_{rom} / 150$  for BCC;  $G_{rom} / 350$  for FCC).
- $M$  = Polycrystalline Taylor factor (2.73 for BCC; 3.06 for FCC).
- $\tau_{ss}$  = Solid-solution strengthening increment:  
 $\tau_{ss} = 0.05 * G_{rom} * (\Delta)^{(2/3)}$
- $G_{rom}$  = Rule-of-mixtures average shear modulus.

### 3. Results & Discovered Candidate Alloys

Genetic algorithm inverse design adhering to the formal HEA constraint (5% to 35 atomic % per element) identified the following optimal alloys:

| Category        | Crystal System | Discovered Composition (at.%)                       | Density (g/cm <sup>3</sup> ) | Yield Strength ( $\sigma_y$ ) | Formation Energy ( $\Delta E_f$ ) |
|-----------------|----------------|-----------------------------------------------------|------------------------------|-------------------------------|-----------------------------------|
| **Refractory**  | BCC (54-atom)  | W:18.5% - Mo:24.7% - Ta:26.1% - Nb:16.8% - V:13.9%  | 14.29                        | **1.96 GPa (1960 MPa)**       | +0.0286 eV/atom (+2.76 kJ/mol)    |
| **Corrosion**   | FCC (48-atom)  | Co:8.6% - Cr:35.4% - Fe:33.1% - Ni:11.2% - Cu:11.6% | 8.00                         | **0.74 GPa (740 MPa)**        | +0.1325 eV/atom                   |
| **Lightweight** | FCC (48-atom)  | Al:35.7% - Mg:11.5% - Li:5.1% - Ti:35.7% - Zn:11.9% | 3.54                         | **0.63 GPa (630 MPa)**        | +0.0966 eV/atom                   |
| **Aerospace**   | FCC (48-atom)  | Al:5.9% - Ti:32.2% - Sc:6.9% - Zr:35.0% - V:20.1%   | 5.01                         | **0.83 GPa (830 MPa)**        | +0.2126 eV/atom                   |

### 4. Proposed Investigation: DFT Validation & Experimental Synthesis

#### 4.1 Ab-Initio Density Functional Theory (Tier 3)

- **Objective:** Establish quantum ground-truth total energies and elastic constants for relaxed SQS supercells.
- **Methodology:** Plane-wave DFT calculations via VASP or Quantum ESPRESSO using Projector Augmented-Wave (PAW) pseudopotentials under Generalized Gradient Approximation (GGA-PBEsol). A plane-wave kinetic energy cutoff of 520 eV and Monkhorst-Pack k-point meshes with a density of 0.03 1/Å will be enforced.
- **Elastic Tensor Derivation:** Strains of +/- 1% and +/- 2% will be applied to compute the elastic stiffness matrix (C11, C12, C44), from which the Voigt-Reuss-Hill polycrystalline bulk modulus (B), shear modulus (G), Young's modulus (E), and Pugh's ductility ratio (B/G) will be evaluated.

#### 4.2 Vacuum Arc Remelting & Experimental Characterization

- **Synthesis:** Ingot synthesis of the top candidate (W-Mo-Ta-Nb-V) via vacuum non-consumable arc melting under ultra-pure Argon (99.999%). Ingots will be flipped and remelted a minimum of seven times to ensure chemical homogeneity, followed by homogenization annealing in a high-vacuum tube furnace (1200°C for 24 h).

- **X-Ray Diffraction (XRD):** Rigaku SmartLab diffractometer (Cu-K<sub>alpha</sub> radiation) to verify single-phase BCC solid-solution crystallinity and confirm the absence of secondary intermetallic reflections.
- **Electron Microscopy (SEM-EDS):** JEOL field-emission scanning electron microscope with energy-dispersive X-ray spectroscopy mapping to analyze microstructural dendrite segregation and partition coefficients.
- **Mechanical Testing:** Microhardness testing (Vickers diamond indenter, 500 gf load, 15 s dwell) across a 10 x 10 indentation matrix to validate predicted yield strengths via the empirical Tabor relation (Yield Strength  $\approx$  Hardness / 3<sup>1/2</sup>).

## 5. Compute Resources, Timeline, and Strategic Value

### 5.1 Resource Allocation

| Resource                             | Requirement         | Technical Justification                                                                    |
|--------------------------------------|---------------------|--------------------------------------------------------------------------------------------|
| **High-Performance Computing (HPC)** | 100,000 core-hours  | High-accuracy ab-initio DFT relaxations and elastic tensor runs across 54-atom supercells. |
| **Software Infrastructure**          | VASP 6.x / QE 7.x   | Validated PAW pseudopotentials for transition metals and refractory species.               |
| **Arc Melting & Metrology**          | KNUST / Partner Lab | Non-consumable tungsten arc furnace, inert atmosphere box, Rigaku XRD, FE-SEM.             |

### 5.2 Project Timeline (6 Months)

|                                                                                     |
|-------------------------------------------------------------------------------------|
| Month 1: Ab-Initio Quantum DFT on SQS Supercells (C11, C12, C44, E <sub>0</sub> )   |
| Month 2: Active Learning Retraining (Interfacing DFT outputs with MetaForge models) |
| Month 3: Vacuum Arc Melting Synthesis of W-Mo-Ta-Nb-V Refractory Ingot              |
| Month 4: XRD Phase Identification & SEM-EDS Chemical Homogeneity Mapping            |
| Month 5: Vickers Microhardness & High-Temperature Oxidation Assessment              |
| Month 6: Manuscript Preparation & Submission to Computational Materials Science     |

## 6. References

1. Yeh, J. W., et al. (2004). "Nanostructured high-entropy alloys with multiple principal elements: novel alloy design concepts and outcomes." *\*Advanced Engineering Materials\**, 6(5), 299-303.
2. Cantor, B., et al. (2004). "Microstructural development in equiatomic multicomponent alloys." *\*Materials Science and Engineering: A\**, 375, 213-218.
3. Senkov, O. N., et al. (2011). "Mechanical properties of Nb<sub>25</sub>Mo<sub>25</sub>Ta<sub>25</sub>W<sub>25</sub> and V<sub>20</sub>Nb<sub>20</sub>Mo<sub>20</sub>Ta<sub>20</sub>W<sub>20</sub> refractory high entropy alloys." *\*Intermetallics\**, 19(5), 698-706.
4. Miracle, D. B., & Senkov, O. N. (2017). "A critical review of high entropy alloys and related concepts." *\*Acta Materialia\**, 122, 448-511.

5. Varvenne, C., Luque, A., & Curtin, W. A. (2016). "Theory of strengthening in dilute and concentrated solid-solution alloys." *\*Acta Materialia\**, 118, 164-176.
6. Takeuchi, A., & Inoue, A. (2005). "Classification of bulk metallic glasses by atomic size difference, heat of mixing and period of constituents." *\*Materials Transactions\**, 46(12), 2817-2829.
7. Deng, B., et al. (2023). "CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling." *\*Nature Machine Intelligence\**, 5(9), 1031-1041.
8. Ward, L., et al. (2018). "Matminer: An open-source toolkit for materials data mining." *\*Computational Materials Science\**, 152, 60-69.
9. Zunger, A., et al. (1990). "Special quasirandom structures." *\*Physical Review Letters\**, 65(3), 353.